ICLES'

MOTILAL JHUNJHUNWALA COLLEGE OF

ARTS, SCIENCE & COMMERCE

Sector 9 A, Vashi, Navi Mumbai – 400 703

Certificate

This is to certify that the project entitled

Industrial IoT Monitoring System for Real-Time Machine Health Monitoring

Undertaken by

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during the academic year 2025-26. In partial fulfilment of B.ScCS (Computer Science) Examination.

It is further certified that he has completed all required phases of the project as given by University of Mumbai.

Internal Guide IT&CS Coordinator

External Examiner Principal

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Lastly, I would like to thank all those who directly and indirectly helped in completion of this project.

Date: Completed by,

Place: Vashi, Navi Mumbai **RUPESH SINGH**

ABSTRACT

The Industrial IoT Monitoring System for Real-Time Machine Health Monitoring is designed to continuously monitor critical machine and environmental parameters using multiple sensors integrated with an ESP8266 NodeMCU microcontroller. The system monitors temperature, humidity, gas leakage, machine vibration, and electrical current consumption in real time to ensure efficient and safe machine operation.

The system utilizes a DHT11 Sensor for temperature and humidity monitoring, an MQ135 Sensor for gas detection, an SW420 Sensor for vibration monitoring, and an ACS712 Sensor for current measurement. All sensor data is processed by the ESP8266 NodeMCU and displayed on a 16×2 LCD display. Whenever any monitored parameter exceeds its predefined threshold value, the system generates immediate visual and audible alerts through an LED indicator and buzzer.

The primary objective of the project is to improve industrial safety, reduce equipment downtime, and support predictive maintenance by identifying abnormal machine conditions at an early stage. Continuous monitoring enables operators to take timely corrective actions before minor issues develop into major failures.

The system also utilizes the built-in Wi-Fi capability of the ESP8266, making it suitable for future Internet of Things (IoT) applications such as cloud monitoring, remote access, and mobile-based supervision. The modular design allows easy integration of additional sensors and advanced monitoring features.

For testing and demonstration purposes, the complete monitoring system was implemented on a portable prototype platform, allowing easy observation of sensor readings and system functionality. The project successfully demonstrates the practical application of IoT technology in industrial monitoring and machine health management.

Overall, the Industrial IoT Monitoring System provides a reliable, cost-effective, and scalable solution for real-time machine health monitoring, industrial safety enhancement, and future smart manufacturing applications.

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INTRODUCTION

1.1 Background

Industrial environments depend heavily on machines for manufacturing, production, packaging, processing, and transportation operations. The performance and reliability of these machines directly affect productivity, operational efficiency, and overall industrial output. Unexpected machine failures can cause production losses, increased maintenance costs, equipment damage, and safety risks for workers.

Traditionally, machine health is monitored through periodic manual inspections. These inspections require human intervention and often fail to identify problems at an early stage. As industries continue to adopt automation and smart manufacturing practices, there is a growing need for systems capable of monitoring machine health continuously and providing immediate alerts whenever abnormal conditions are detected.

The Internet of Things (IoT) has emerged as a transformative technology that enables devices, sensors, and machines to communicate and exchange data over a network. By integrating sensors with microcontrollers and wireless communication technologies, IoT systems can collect real-time information from industrial equipment and provide continuous monitoring without human intervention.

Machine health monitoring is one of the most important applications of IoT technology. Parameters such as temperature, humidity, vibration, air quality, and electrical current can be monitored continuously to identify early signs of equipment malfunction. Real-time analysis of these parameters helps industries implement predictive maintenance strategies, reduce downtime, and improve workplace safety.

The Industrial IoT Monitoring System for Real-Time Machine Health Monitoring is developed to provide an efficient and cost-effective solution for monitoring critical machine parameters. The system utilizes an ESP8266 NodeMCU microcontroller connected to multiple sensors including DHT11, MQ135, SW420, and ACS712. Sensor readings are processed continuously and displayed on an LCD screen. Whenever abnormal conditions are detected, the system activates visual and audible warning mechanisms using LEDs and a buzzer.

For prototype development and demonstration purposes, the monitoring unit is mounted on a mobile RC chassis platform. This allows easy transportation, testing, and practical demonstration of the monitoring system while maintaining the core objective of industrial machine health monitoring.

The project demonstrates how modern IoT technology can be used to improve industrial safety, reduce maintenance costs, minimize unexpected failures, and support smart manufacturing environments.

1.2 Objectives

The main objectives of the Industrial IoT Monitoring System are:

- To continuously monitor machine temperature.
- To monitor humidity levels in the operating environment.
- To detect harmful gas leakage using an air quality sensor.
- To monitor machine vibration and identify abnormal mechanical activity.
- To measure electrical current consumption of machinery.
- To display real-time sensor readings on an LCD display.
- To provide immediate visual alerts using LEDs.
- To generate audible alerts using a buzzer during abnormal conditions.
- To support preventive and predictive maintenance practices.
- To improve industrial safety and operational efficiency.
- To reduce machine downtime and maintenance costs.
- To demonstrate practical implementation of Industrial Internet of Things technology.

1.3 Purpose, Scope and Applicability

1.3.1 Purpose

The primary purpose of developing the Industrial IoT Monitoring System is to provide a reliable and cost-effective solution for real-time machine health monitoring. The system continuously monitors important machine parameters such as temperature, humidity, air quality, electrical current, and vibration in order to detect potential faults before they become serious failures.

Another purpose of the project is to demonstrate the practical application of IoT technology in industrial environments. By integrating multiple sensors with an ESP8266 microcontroller, the system enables continuous monitoring and instant alert generation.

The project also aims to:

- Improve machine reliability.
- Enhance worker safety.
- Reduce maintenance expenses.
- Prevent unexpected equipment failures.
- Demonstrate real-time industrial monitoring techniques.
- Support Industry 4.0 concepts and smart manufacturing applications.

1.3.2 Scope

The scope of the Industrial IoT Monitoring System covers real-time machine condition monitoring using multiple sensors and embedded systems technology.

From a technical perspective, the project includes:

- Temperature Monitoring
- Humidity Monitoring
- Gas Leakage Detection
- Vibration Detection
- Current Monitoring
- LCD-Based Data Visualization
- LED and Buzzer Alert System
- Wi-Fi Enabled IoT Communication through ESP8266

The developed system can be implemented in:

- Manufacturing Industries
- Production Plants
- Warehouses
- Mechanical Workshops
- Research Laboratories
- Processing Industries

The system architecture also allows future enhancements such as:

- Cloud Data Storage
- Mobile Application Monitoring
- SMS Notification System
- Machine Learning Based Fault Prediction
- IoT Dashboard Integration
- Remote Monitoring through Web Platforms

Thus, the project provides a scalable foundation for future industrial automation and predictive maintenance applications.

1.3.3 Applicability

The Industrial IoT Monitoring System can be applied in various sectors where continuous equipment monitoring is required.

1. Industrial Applications

- Machine Health Monitoring
- Production Line Monitoring
- Manufacturing Equipment Monitoring
- Safety Surveillance

2. Power and Electrical Systems

- Current Monitoring
- Electrical Load Analysis
- Equipment Protection

3. Safety Monitoring

- Gas Leakage Detection
- Environmental Monitoring
- Hazard Detection

4. Maintenance Applications

- Predictive Maintenance
- Preventive Maintenance
- Fault Identification

5. Educational and Research Use

- IoT Training Projects
- Embedded System Research
- Industrial Automation Demonstrations
- Academic Learning Platforms

Because of its simplicity, low cost, and scalability, the system is suitable for both educational and real-world industrial applications.

1.4 Achievements

The successful development and implementation of the Industrial IoT Monitoring System resulted in several important technical achievements.

1. Successful IoT Integration

The project successfully integrates multiple sensors with the ESP8266 microcontroller, demonstrating the effective implementation of IoT technology for industrial monitoring.

2. Real-Time Machine Monitoring

The system continuously monitors machine conditions and provides live updates through the LCD display.

3. Multi-Sensor Data Acquisition

The project successfully acquires data from:

- DHT11 Temperature Sensor
- DHT11 Humidity Sensor
- MQ135 Gas Sensor
- SW420 Vibration Sensor
- ACS712 Current Sensor

4. Alert Generation System

An effective warning mechanism has been implemented using:

- LED Indicators
- Active Buzzer
- LCD Warning Messages

5. Fault Detection Capability

The system can detect:

- High Temperature
- High Humidity
- Gas Leakage
- Excessive Current Consumption
- Abnormal Machine Vibrations

6. LCD-Based User Interface

Sensor readings and warning messages are displayed on a 16×2 LCD display, providing a user-friendly monitoring interface.

7. Portable Prototype Development

The complete system was successfully integrated onto a prototype testing platform using an RC chassis, enabling convenient testing and demonstration.

8. Low-Cost Industrial Monitoring Solution

The project demonstrates that reliable industrial monitoring can be achieved using affordable sensors and open-source development tools.

Overall Achievement Summary

The Industrial IoT Monitoring System successfully fulfills its objective of monitoring machine health in real time. The integration of IoT technology, sensor networks, embedded systems, and alert mechanisms provides an efficient solution for industrial safety and equipment monitoring. The project establishes a strong foundation for future smart industrial applications and predictive maintenance systems.

SURVEY OF TECHNOLOGIES

2.1 Awareness and Understanding of Available Technologies

Before developing the Industrial IoT Monitoring System, various technologies used in industrial monitoring and automation were studied and analyzed. Understanding available technologies helped in selecting the most suitable, reliable, and cost-effective solution for the project.

1. Manual Monitoring Systems

Traditional industries often rely on manual monitoring of machinery. Operators periodically inspect equipment and record observations manually.

Advantages:

- Simple implementation
- No specialized hardware required

Limitations:

- Time-consuming
- Human errors
- Delayed fault detection
- High maintenance effort

Due to these limitations, automated monitoring systems are preferred in modern industries.

2. GSM-Based Monitoring Systems

GSM technology enables remote monitoring through SMS notifications and cellular communication networks.

Advantages:

- Long-distance communication
- SMS-based alerts

Limitations:

- Additional operational cost
- Network dependency
- Slow response time

3. Wi-Fi Based IoT Systems

Wi-Fi based monitoring systems utilize internet connectivity for real-time data transfer and remote access.

Advantages:

- Real-time data monitoring
- Internet connectivity
- Remote access capability
- Cloud integration support

Limitations:

- Requires stable internet connection
- Network configuration required

4. Arduino-Based Monitoring Systems

Arduino boards are widely used in embedded applications for sensor monitoring and automation.

Advantages:

- Easy programming
- Large community support
- Multiple sensor compatibility

Limitations:

- Additional Wi-Fi module required
- Lower network integration capability

5. ESP8266 Based IoT Systems

ESP8266 NodeMCU combines microcontroller functionality with built-in Wi-Fi connectivity.

Advantages:

- Built-in Wi-Fi
- Low power consumption
- Cost effective
- Easy programming
- Excellent IoT capability
- Multiple sensor support

Limitations:

- Limited GPIO pins compared to some controllers

After studying the available technologies, ESP8266 was selected because it provides built-in Wi-Fi connectivity, easy sensor interfacing, low cost, and excellent support for IoT applications.

2.2 Technologies Used

Based on the survey and analysis, the following technologies were implemented in the Industrial IoT Monitoring System.

1. ESP8266 NodeMCU

ESP8266 NodeMCU acts as the main controller of the system.

Functions:

- Reads sensor data
- Processes sensor values
- Detects abnormal conditions
- Controls LCD display
- Activates buzzer and LED alerts
- Provides Wi-Fi capability

2. DHT11 Temperature and Humidity Sensor

The DHT11 sensor is used to measure environmental temperature and humidity.

Functions:

- Temperature measurement
- Humidity measurement
- Real-time environmental monitoring

Benefits:

- Low cost
- Easy interfacing
- Reliable performance

3. MQ135 Gas Sensor

The MQ135 sensor is used for air quality monitoring and gas leakage detection.

Functions:

- Detect hazardous gases

- Monitor air quality
- Industrial safety monitoring

Benefits:

- Fast response
- Good sensitivity
- Low cost implementation

4. SW420 Vibration Sensor

The SW420 sensor detects machine vibration.

Functions:

- Detect abnormal machine vibration
- Indicate mechanical faults
- Monitor machine health

Benefits:

- Simple implementation
- Reliable vibration detection

5. ACS712 Current Sensor

The ACS712 sensor measures electrical current consumption.

Functions:

- Current monitoring
- Load analysis
- Fault detection

Benefits:

- Electrical safety
- Continuous current monitoring

6. LCD Display with I2C Module

A 16×2 LCD display is used to visualize sensor readings and warning messages.

Functions:

- Display sensor values
- Display warning alerts
- Provide user interface

Benefits:

- Simple operation
- Easy readability
- Low power consumption

7. LED Alert System

The LED serves as a visual warning indicator.

Functions:

- Visual fault indication
- Alarm notification

8. Buzzer Alert System

The buzzer generates audible warning signals.

Functions:

- Audio alerts
- Immediate fault notification

9. Wi-Fi Communication Technology

ESP8266 provides built-in wireless communication capabilities.

Functions:

- IoT connectivity
- Remote monitoring support
- Future cloud integration

2.3 Program Code

The Industrial IoT Monitoring System is developed using Arduino IDE and Embedded C/C++ programming language.

The software program performs the following operations:

Sensor Reading

The ESP8266 continuously reads data from:

- DHT11 Temperature Sensor
- DHT11 Humidity Sensor
- MQ135 Gas Sensor

- SW420 Vibration Sensor
- ACS712 Current Sensor

Data Processing

The microcontroller processes sensor values and compares them with predefined threshold limits.

Examples:

- High Temperature Detection
- High Humidity Detection
- Gas Leakage Detection
- High Current Detection
- Abnormal Vibration Detection

LCD Display Management

The LCD screen displays:

- Temperature Values
- Humidity Values
- Gas Sensor Status
- Current Sensor Readings
- Vibration Status

Alert System

When abnormal readings are detected:

- LED starts blinking
- Buzzer activates
- Warning message appears on LCD

Example:

Plain Text

*** WARNING ***

HIGH TEMP

Program Advantages

- Real-time monitoring
- Automatic fault detection
- User-friendly display
- Fast response system
- Reliable industrial monitoring

The Arduino program acts as the core intelligence of the Industrial IoT Monitoring System and enables continuous monitoring of machine health parameters while ensuring timely fault detection and alert generation.

REQUIREMENTS AND ANALYSIS

3.1 Problem Definition

In today's industrial environment, machines are continuously used in manufacturing, production, processing, and assembly operations. Unexpected machine failures can lead to production losses, equipment damage, increased maintenance costs, and safety hazards. Traditional machine monitoring methods often depend on manual inspection and periodic maintenance, which may fail to detect problems at an early stage.

Machine health is influenced by various parameters such as temperature, humidity, vibration, gas leakage, and electrical current consumption. Any abnormal change in these parameters can indicate machine malfunction or unsafe operating conditions.

Therefore, there is a need for an automated and intelligent monitoring system capable of continuously observing machine conditions in real time and generating alerts whenever abnormal situations occur.

The proposed Industrial IoT Monitoring System for Real-Time Machine Health Monitoring addresses this problem by integrating multiple sensors with an ESP8266 NodeMCU microcontroller. The system continuously monitors machine parameters, displays data on an LCD screen, and provides warning alerts through LEDs and a buzzer whenever unsafe conditions are detected.

3.2 Requirement Specification

The Industrial IoT Monitoring System is designed to monitor machine health parameters continuously and provide timely alerts in the event of abnormal conditions. The requirements are classified into Functional Requirements and Non-Functional Requirements.

A. Functional Requirements

Functional requirements define the primary operations performed by the system.

1. Temperature Monitoring

- The system must continuously monitor temperature using the DHT11 sensor.
- Temperature readings must be displayed on the LCD display.
- Alerts must be generated whenever temperature exceeds the predefined threshold.

2. Humidity Monitoring

- The system must continuously monitor humidity levels.
- Humidity readings must be displayed in real time.
- Warning notifications must be generated during abnormal humidity conditions.

3. Gas Leakage Detection

- The system must detect harmful gases using the MQ135 sensor.
- Gas leakage conditions must activate warning alerts.
- Visual and audible indications must be generated immediately.

4. Vibration Monitoring

- The SW420 vibration sensor must monitor machine vibration.
- Excessive vibration conditions must generate warning alerts.
- The system must help identify potential mechanical faults.

5. Current Monitoring

- The ACS712 sensor must measure current consumption.
- Excessive current conditions must be detected.
- Alerts must be generated whenever abnormal current readings occur.

6. LCD Monitoring System

- Sensor readings must be displayed on a 16×2 LCD display.
- Warning messages must be shown when abnormal conditions are detected.
- Display information must update automatically.

7. Alert System

- LED indicators must provide visual alerts.
- A buzzer must generate audible alerts.
- Alerts must notify the operator immediately during fault conditions.

B. Non-Functional Requirements

1. Reliability

- The system must operate continuously without interruption.
- Sensors must provide stable performance.
- Fault detection should be accurate and dependable.

2. Performance

- The system must provide real-time monitoring.
- Response time should be minimal.
- Data processing must be efficient.

3. Accuracy

- Sensor readings should be reasonably accurate.
- False alarms should be minimized.

4. Maintainability

- Components should be easy to replace.
- The system should support future upgrades.

5. Cost Effectiveness

- The system should use affordable and readily available components.
- Operational and maintenance costs should be low.

C. Operational Requirements

- Controller: ESP8266 NodeMCU
- Temperature Sensor: DHT11
- Humidity Sensor: DHT11
- Gas Sensor: MQ135
- Vibration Sensor: SW420
- Current Sensor: ACS712
- Display Module: 16×2 LCD with I2C
- Alert Devices: LED and Buzzer
- Power Supply: 5V DC
- Communication: Wi-Fi

3.3 Planning and Scheduling

Effective planning and scheduling were necessary to ensure the successful development and implementation of the Industrial IoT Monitoring System. The project was divided into several phases to achieve systematic execution.

1. Project Planning Phase

The following activities were performed during the planning stage:

- Understanding project objectives.
- Defining system requirements.
- Selecting appropriate sensors and controller.
- Studying Industrial IoT concepts and applications.

This phase established the project foundation and development objectives.

2. Research and Technology Analysis

During this phase:

- Different IoT technologies were studied.
- Suitable sensors were identified.
- ESP8266 was selected as the main controller.
- Various machine monitoring techniques were analyzed.

This research helped determine the most efficient implementation approach.

3. System Design Planning

Activities included:

- Designing the system architecture.
- Planning sensor integration.
- Designing the LCD monitoring interface.
- Planning alert generation mechanisms.
- Preparing block diagrams and workflow designs.

4. Component Procurement

The required components were arranged:

- ESP8266 NodeMCU
- DHT11 Sensor
- MQ135 Gas Sensor
- SW420 Vibration Sensor
- ACS712 Current Sensor
- LCD Display
- LED
- Buzzer
- Breadboard
- Jumper Wires
- Power Supply

5. Hardware Assembly

This phase involved:

- Connecting sensors to the ESP8266.
- Integrating the LCD display.
- Connecting the buzzer and LED.
- Verifying power supply connections.
- Testing electrical connections.

6. Software Development

This phase included:

- Arduino IDE configuration.
- Sensor programming.
- LCD programming.
- Alarm logic implementation.
- System integration coding.

7. Testing and Debugging

Testing procedures included:

- Temperature sensor testing.
- Humidity sensor testing.
- Gas sensor testing.
- Vibration sensor testing.
- Current sensor testing.
- LCD display testing.
- LED and buzzer testing.

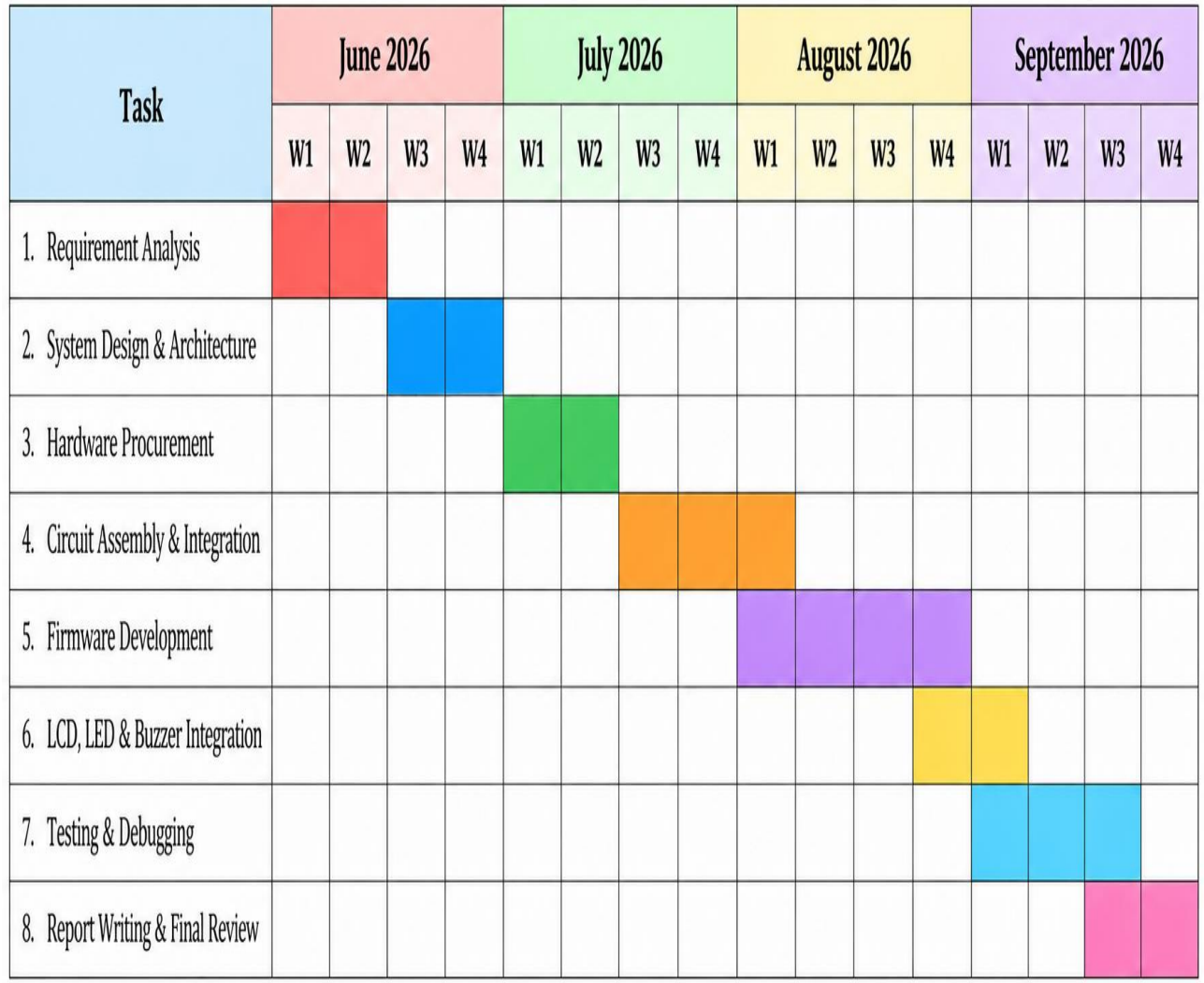
Necessary corrections were made whenever issues were identified.

8. Final Implementation and Optimization

Final improvements included:

- Optimizing sensor readings.
- Improving system stability.
- Verifying alarm functionality.
- Conducting complete system testing.

The Industrial IoT Monitoring System was successfully implemented and tested under different operating conditions.



3.4 Software and Hardware Requirement

A. Software Requirement

The Industrial IoT Monitoring System requires both software and hardware resources for successful implementation and operation.

1. Arduino IDE

Arduino IDE is used for writing, compiling, and uploading the program code to the ESP8266 NodeMCU. It provides an easy-to-use development environment for embedded programming.

Functions:

- Program development
- Code compilation
- Program uploading
- Debugging and testing

2. ESP8266 Board Package

The ESP8266 board package enables Arduino IDE to support NodeMCU programming and library integration.

Functions:

- ESP8266 support
- Firmware uploading
- Library management

3. Serial Monitor

The Serial Monitor is used for debugging and monitoring sensor values during testing.

Functions:

- Display sensor readings
- Monitor system status
- Troubleshooting

4. Microsoft Word

Microsoft Word is used for preparing and formatting the project documentation.

5. Draw.io / Lucidchart

These tools are used for designing UML diagrams, flowcharts, and system architecture diagrams.

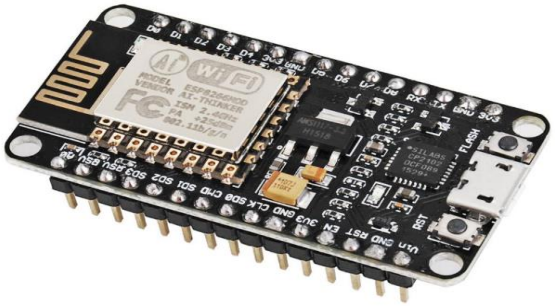
B. Hardware Requirement

1. ESP8266 NodeMCU

The ESP8266 NodeMCU serves as the main controller of the system. It collects sensor data, processes information, activates alerts, and provides Wi-Fi connectivity.

Functions:

- Sensor interfacing
- Data processing
- Alert generation
- IoT communication

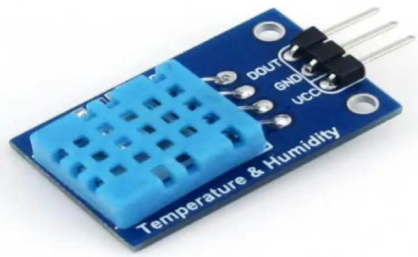


2. DHT11 Sensor

The DHT11 sensor measures environmental temperature and humidity.

Functions:

- Temperature monitoring
- Humidity monitoring
- Environmental condition monitoring



3. MQ135 Gas Sensor

The MQ135 sensor detects harmful gases and monitors air quality.

Functions:

- Gas leakage detection
- Air quality monitoring
- Safety monitoring

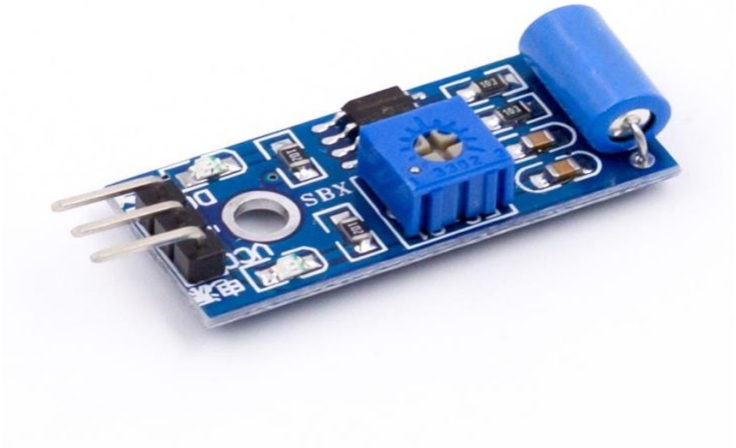


4. SW420 Vibration Sensor

The SW420 sensor detects vibrations and abnormal machine movement.

Functions:

- Vibration detection
- Fault indication
- Machine health monitoring

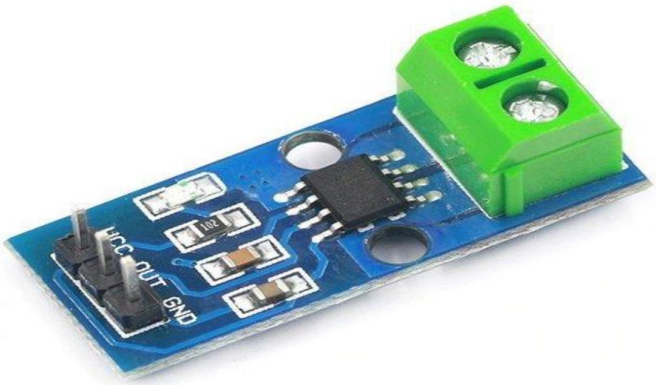


5. ACS712 Current Sensor

The ACS712 sensor measures electrical current consumed by the machine.

Functions:

- Current monitoring
- Power analysis
- Overload detection



6. 16×2 LCD Display with I2C Module

The LCD displays real-time sensor readings and warning messages.

Functions:

- Data display
- Warning notification
- User interface



7. LED Indicator

The LED provides visual warning signals during abnormal conditions.

Functions:

- Visual alert generation
- Fault indication

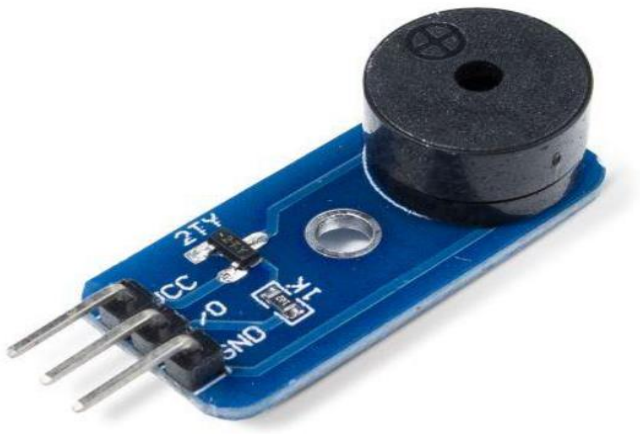


8. Active Buzzer

The buzzer generates audible alarms whenever abnormal conditions are detected.

Functions:

- Audible alerts
- Warning indication

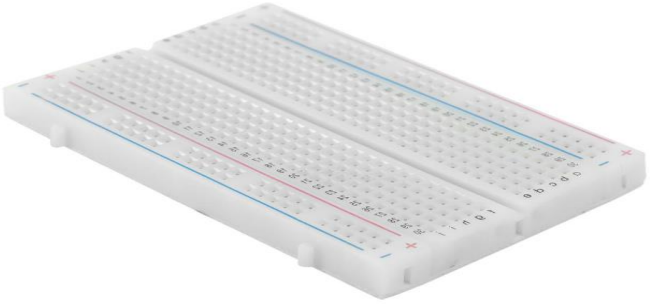


9. Breadboard

The breadboard is used for temporary circuit assembly and testing.

Functions:

- Circuit prototyping
- Component connectivity



10. Jumper Wires

Jumper wires provide electrical connections between components.

Functions:

- Signal transmission
- Power connections



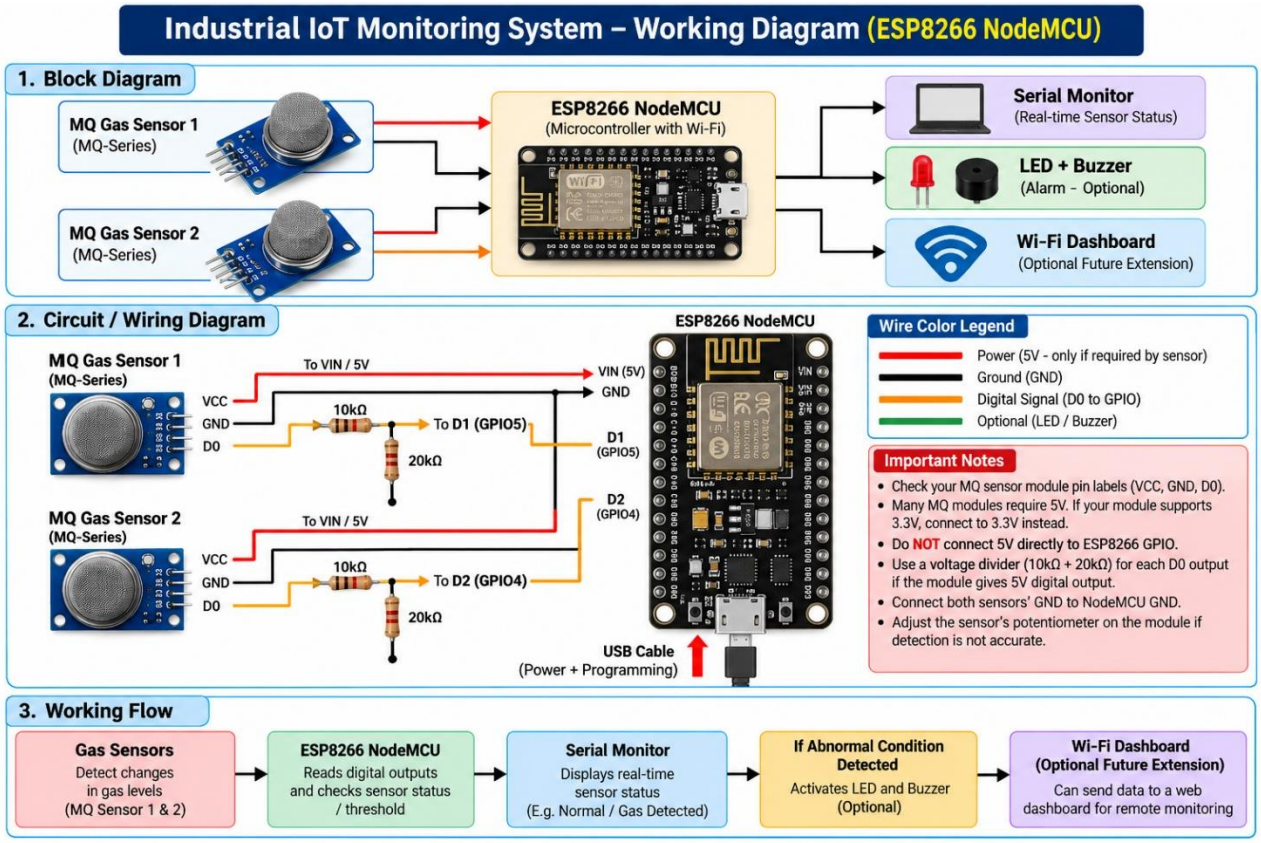
System Integration Overview

The Industrial IoT Monitoring System integrates multiple sensors, display modules, and alert devices with the ESP8266 NodeMCU controller to provide real-time machine health monitoring. The DHT11 sensor measures temperature and humidity, the MQ135 sensor monitors gas levels, the SW420 sensor detects vibrations, and the ACS712 sensor measures current consumption. All sensor data is continuously collected and processed by the ESP8266 NodeMCU.

The processed information is displayed on the 16×2 LCD display for real-time monitoring. Whenever any sensor reading exceeds the predefined threshold value, the ESP8266 activates the LED and buzzer to generate visual and audible warnings. The built-in Wi-Fi capability of the ESP8266 allows future integration with IoT platforms for remote monitoring and data analysis.

The complete system follows the flow:

Sensors → ESP8266NodeMCU → Data Processing → LCD Display → Alert System → IoT Monitoring



SYSTEM DESIGN

The System Design phase defines the overall architecture, components, and working methodology of the Industrial IoT Monitoring System for Real-Time Machine Health Monitoring. It describes how various hardware and software components are integrated to continuously monitor machine conditions and generate alerts whenever abnormal situations occur.

The system is built around the ESP8266 NodeMCU microcontroller, which acts as the central processing unit. Multiple sensors including the DHT11 Temperature and Humidity Sensor, MQ135 Gas Sensor, SW420 Vibration Sensor, and ACS712 Current Sensor are connected to the controller. These sensors continuously collect real-time data related to machine health and environmental conditions.

The ESP8266 receives sensor data, processes the readings, and compares them with predefined threshold values. Under normal operating conditions, all sensor values are displayed sequentially on the 16×2 LCD display, allowing the user to monitor the machine status in real time. When any sensor reading exceeds its safety limit, the system immediately activates the alert mechanism.

The alert mechanism consists of an LED indicator and a buzzer. The LED provides visual indication while the buzzer message indicating the type of fault detected, such as high temperature, gas leakage, excessive vibration, high humidity, or abnormal current consumption.

The system also utilizes the built-in Wi-Fi capability of the ESP8266, making it suitable for future IoT applications such as cloud monitoring, remote access, mobile applications, and smart industrial automation. The modular design allows additional sensors and monitoring features to be incorporated easily without major modifications.

Overall, the Industrial IoT Monitoring System provides continuous machine health monitoring, rapid fault detection, real-time data visualization, and alert generation. This design improves industrial safety, reduces equipment downtime, supports predictive maintenance, and enhances operational efficiency.

4.1 Model Used

The project uses an Internet of Things (IoT)-based monitoring model. It uses two MQ-series gas sensor modules and an ESP8266NodeMCU microcontroller to monitor gas conditions.

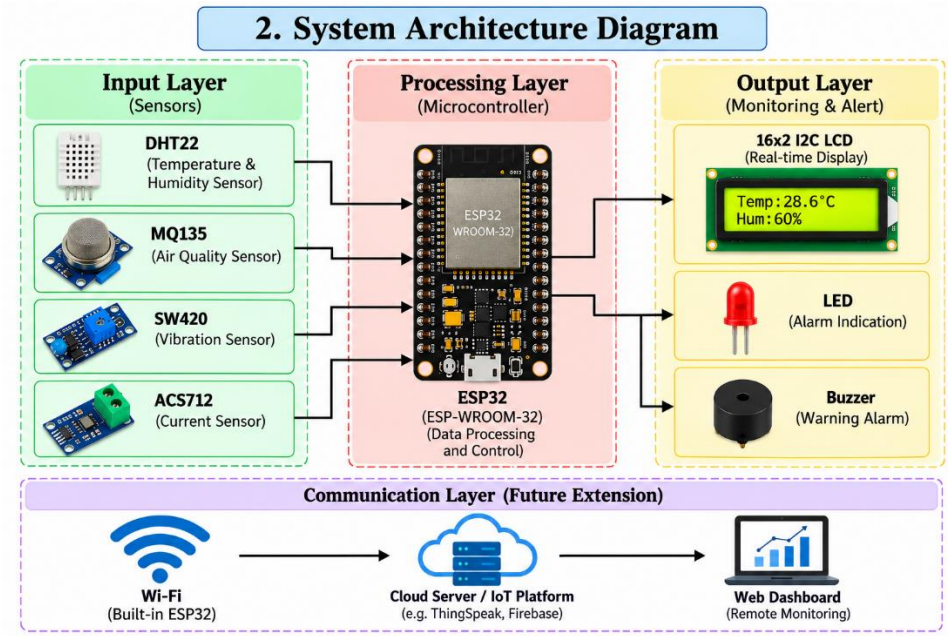
The sensors detect changes in gas levels and send digital signals to the NodeMCU. The microcontroller processes these signals and displays the sensor status on the Serial Monitor. An LED and buzzer can be added to provide alerts when gas is detected.

The main components of the model are:

1. Input Module: Two MQ-series gas sensors.
2. Processing Module: ESP8266NodeMCU.
3. Monitoring Module: Serial Monitor.
4. Alert Module: Optional LED and buzzer.
5. Communication Module: Wi-Fi capability for future remote monitoring.

4.2 System Architecture Diagram

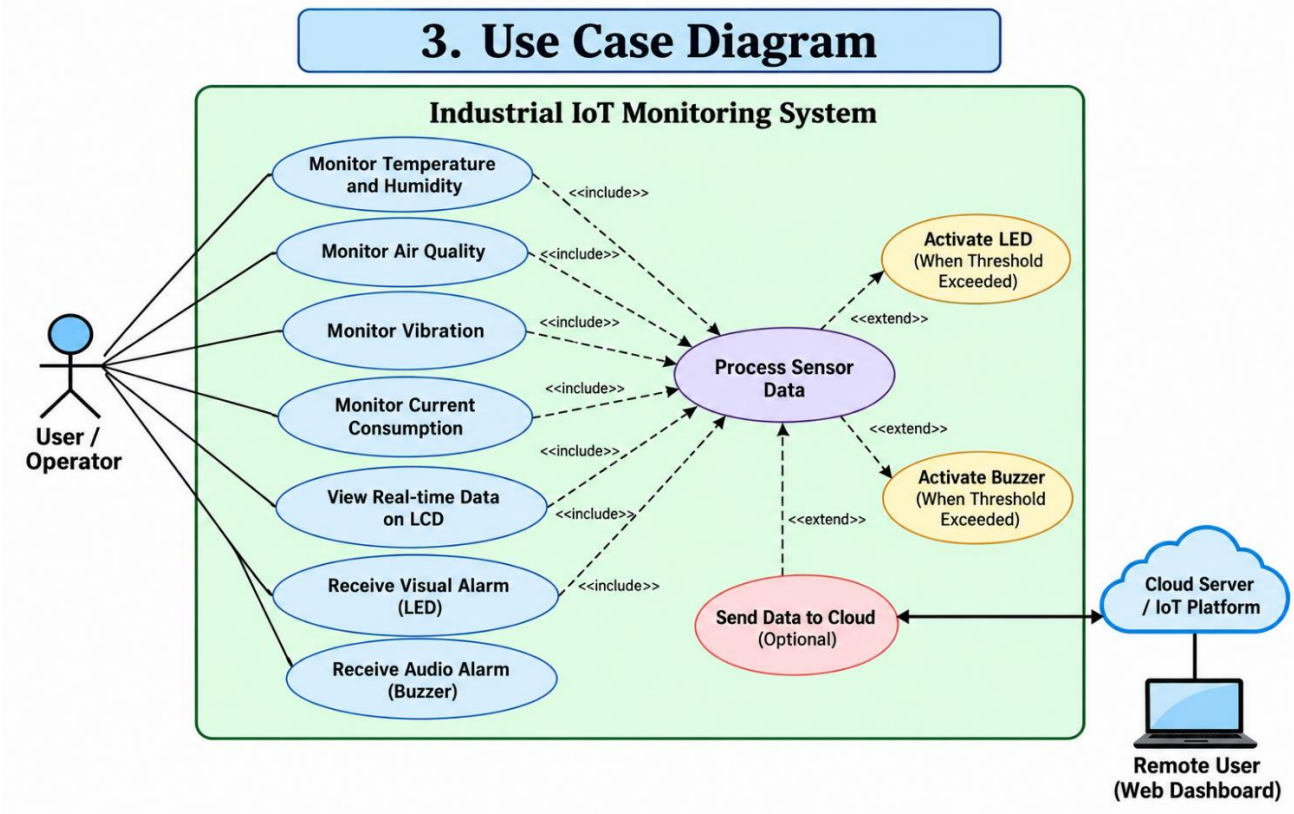
This diagram shows how the components communicate with each other.



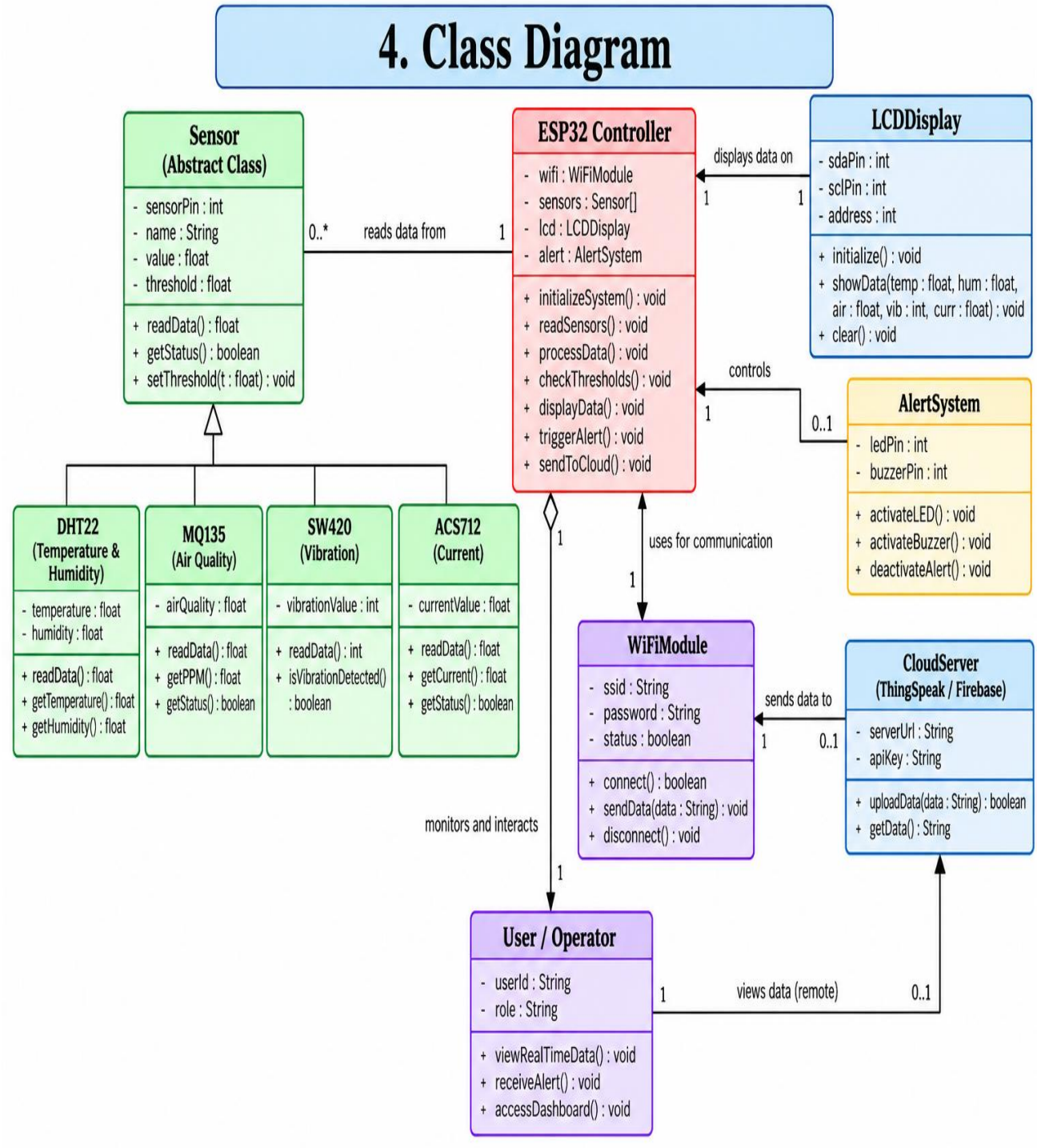
4.3 UML Diagram

The UML diagrams represent the structure, behavior, and workflow of the Industrial IoT Monitoring System

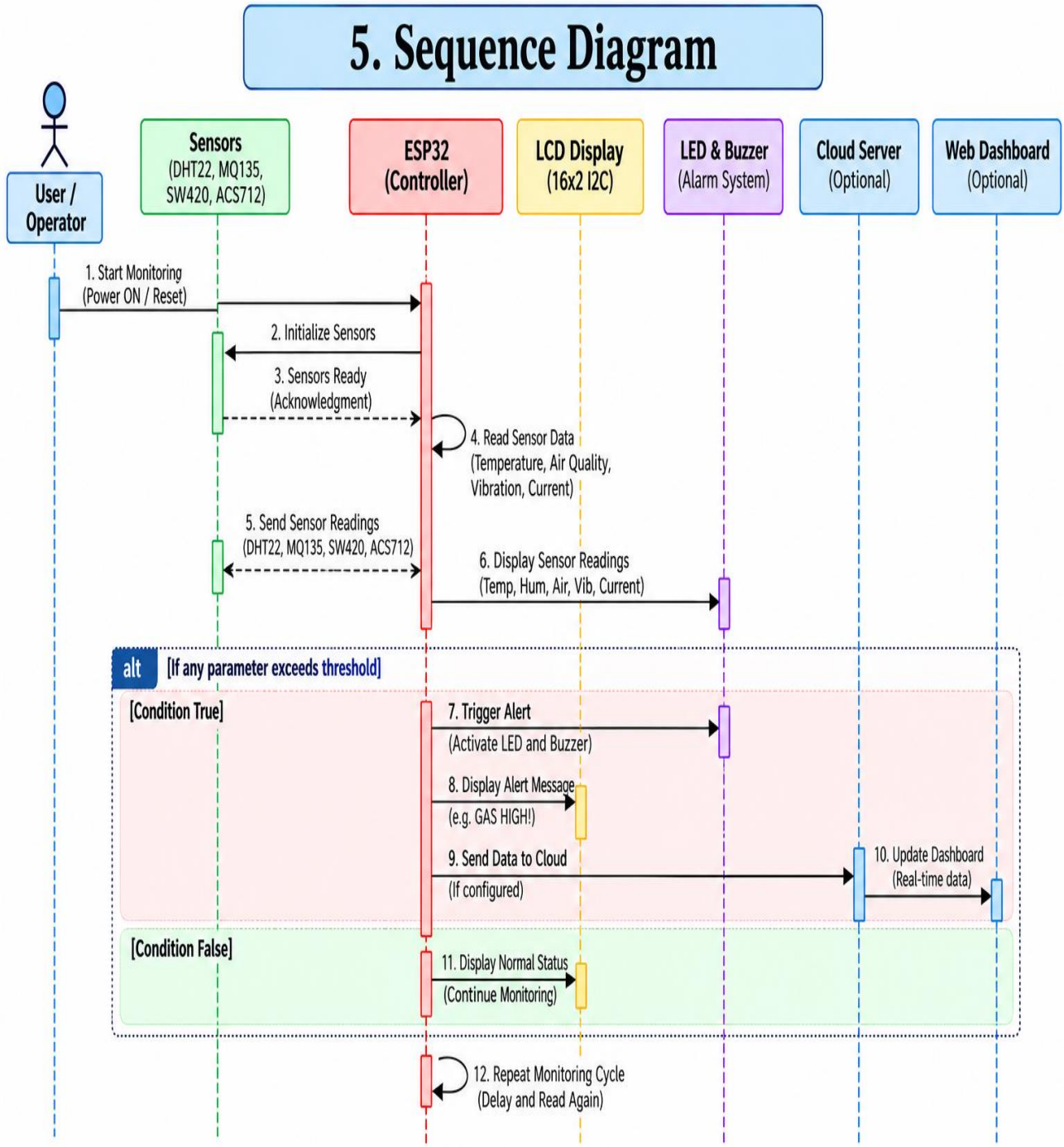
4.3.1 Use Case Diagram



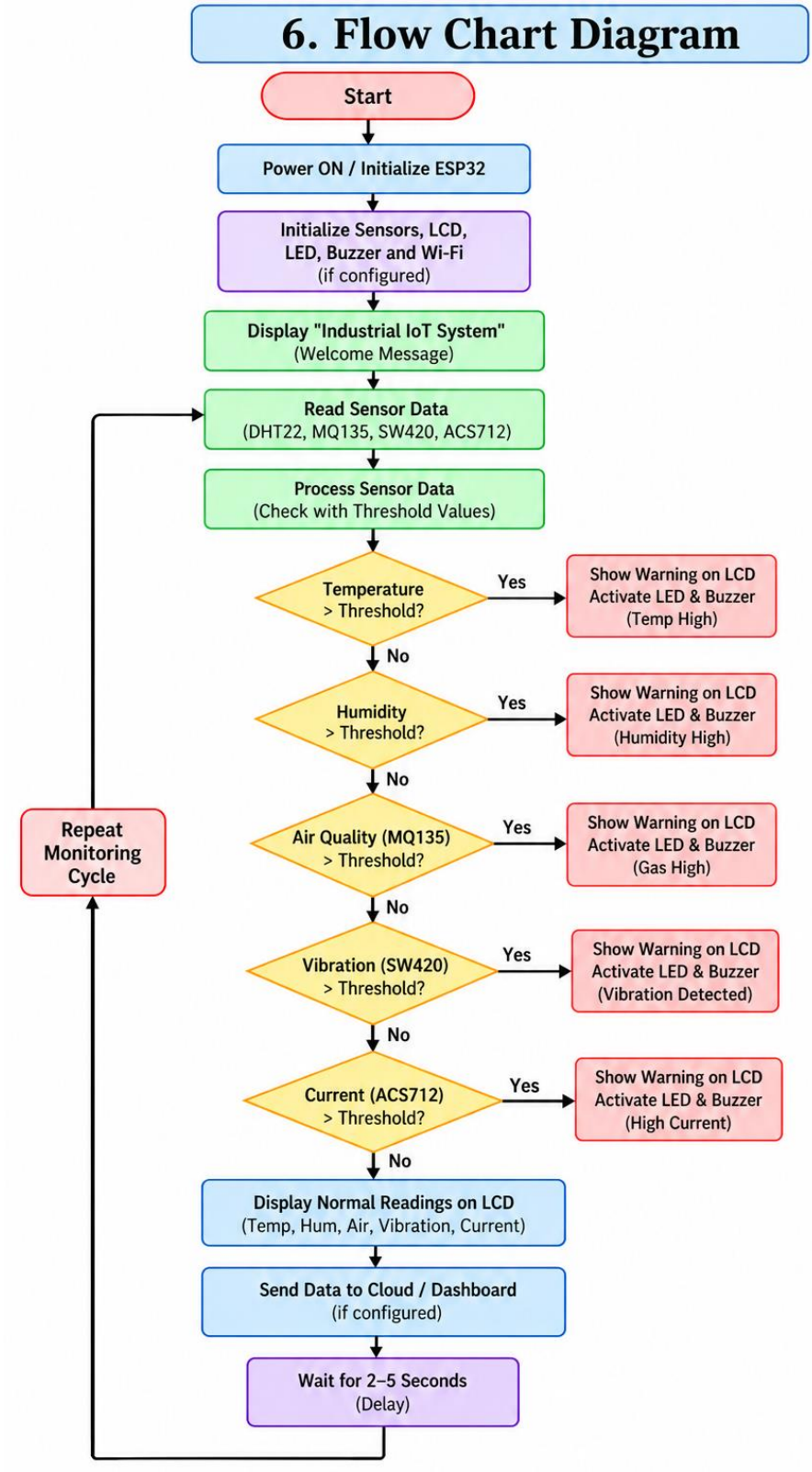
4.3.2 Class Diagram



4.3.3 Sequence Diagram



4.3.4 Flow Chart Diagram



PROJECT CODE

Code of Project

The Industrial IoT Monitoring System is developed using an **ESP8266 NodeMCU** microcontroller and programmed using the **Arduino IDE**. The system continuously monitors important machine health parameters such as temperature, humidity, gas leakage, vibration, and current consumption using different sensors connected to the ESP8266.

The controller continuously receives data from the sensors, processes the readings, and compares them with predefined threshold values. When any abnormal condition is detected, the system activates the LED and buzzer while displaying a warning message on the LCD display.

Working Logic

Although the system uses a software-based control approach, its working can be logically represented as follows:

Plain Text

IF Temperature > Threshold

→ Activate LED

→ Activate Buzzer

→ Display "HIGH TEMP"

IF Humidity > Threshold

→ Activate LED

→ Activate Buzzer

→ Display "HIGH HUMIDITY"

IF Gas Detected

→ Activate LED

→ Activate Buzzer

→ Display "GAS ALERT"

IF Current > Threshold

→ Activate LED

→ Activate Buzzer

→ Display "HIGH CURRENT"

IF Vibration Detected

→ Activate LED

→ Activate Buzzer

→ Display "VIBRATION ALERT"

ELSE

→ Continue Monitoring

→ Display Sensor Data

Features of the Program

- Real-time sensor monitoring.
- Automatic fault detection.
- LCD-based status display.
- Instant visual and audible alerts.
- Continuous monitoring operation.
- Expandable IoT architecture.
- Easy maintenance and modification.

Advantages of the Developed System

- Improves industrial safety.
- Reduces machine downtime.
- Enables early fault detection.
- Supports predictive maintenance.
- Cost-effective implementation.
- Easy to use and operate.
- Reliable monitoring performance.

COMPONENTS & DIAGRAMS

This section describes all the hardware components used in the Industrial IoT Monitoring System along with the circuit and block diagram representation. The system is designed to continuously monitor machine health parameters and generate warning alerts whenever abnormal conditions are detected.

6.1 Components Used in the Project

The following components are used in the Industrial IoT Monitoring System:

1. ESP8266NodeMCU

The ESP8266NodeMCU serves as the main controller of the system. It collects sensor data, processes the information, controls the LCD display, and activates the alert system. The built-in Wi-Fi capability allows future IoT integration and remote monitoring applications.

Functions:

- Sensor interfacing
- Data processing
- Wi-Fi communication
- Alert generation

2. DHT11 Temperature and Humidity Sensor

The DHT11 sensor is used to measure environmental temperature and humidity. It provides real-time readings that help monitor machine operating conditions and surrounding environmental parameters.

Functions:

- Temperature monitoring
- Humidity monitoring
- Environmental analysis

3. MQ135 Gas Sensor

The MQ135 sensor is used to detect harmful gases and monitor air quality. It improves industrial safety by identifying gas leakage and abnormal air conditions.

Functions:

- Gas leakage detection
- Air quality monitoring
- Safety monitoring

4. SW420 Vibration Sensor

The SW420 sensor detects vibration generated by machine operation. Excessive vibration can indicate mechanical problems, imbalance, or equipment malfunction.

Functions:

- Vibration monitoring
- Machine fault detection
- Equipment health analysis

5. ACS712 Current Sensor

The ACS712 sensor measures electrical current consumed by the machine. It helps identify overload conditions and abnormal current consumption.

Functions:

- Current measurement
- Overload detection
- Electrical monitoring

6. 16×2 LCD Display with I2C Module

The LCD display provides a user-friendly interface for displaying real-time sensor readings and warning messages.

Functions:

- Sensor data display
- Status monitoring
- Warning indication

7. LED Indicator

The LED provides visual alerts whenever abnormal machine conditions are detected.

Functions:

- Visual warning generation
- Fault indication

8. Active Buzzer

The buzzer generates audible warning signals when sensor readings exceed predefined threshold values.

Functions:

- Audible alerts
- Alarm generation

9. Breadboard

The breadboard is used for assembling and testing the prototype without soldering.

Functions:

- Circuit prototyping
- Component interconnection

10. Jumper Wires

Jumper wires establish electrical connections between sensors, controller, and peripheral devices.

Functions:

- Signal transmission
- Power distribution

6.2 Circuit Diagram

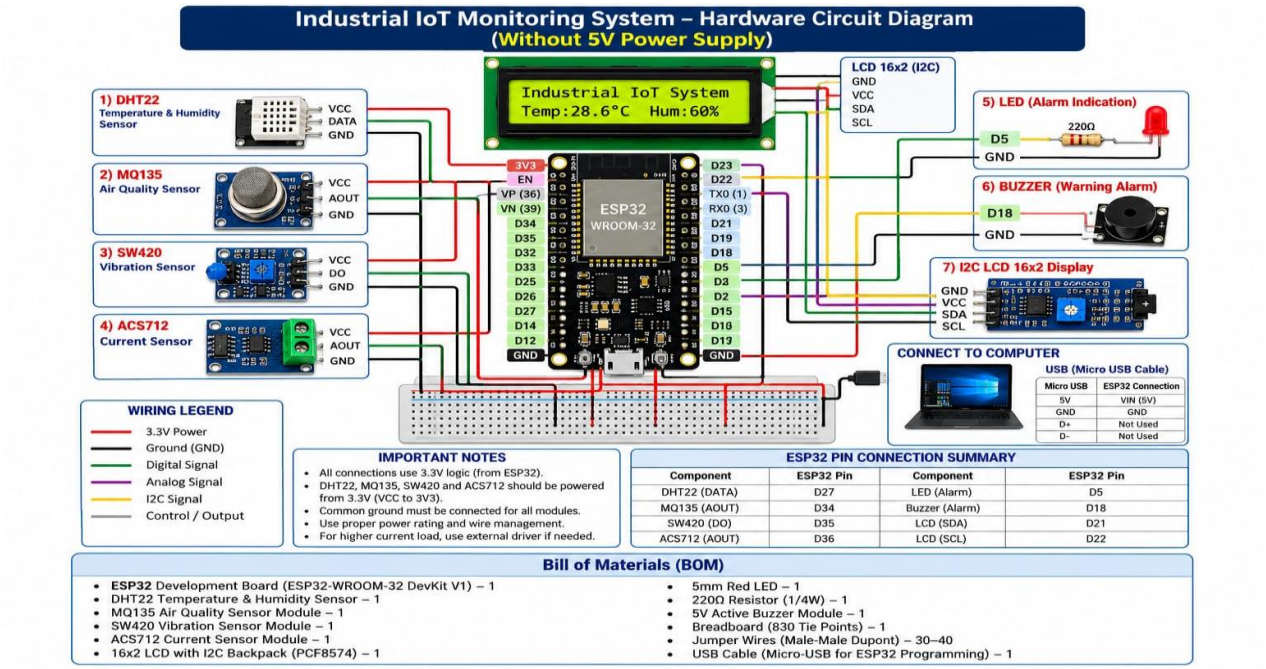
Circuit Description

The circuit consists of an ESP8266 NodeMCU connected to multiple sensors including DHT11, MQ135, SW420, and ACS712. The LCD display is connected through the I2C interface for displaying sensor values. An LED and buzzer are connected as warning devices.

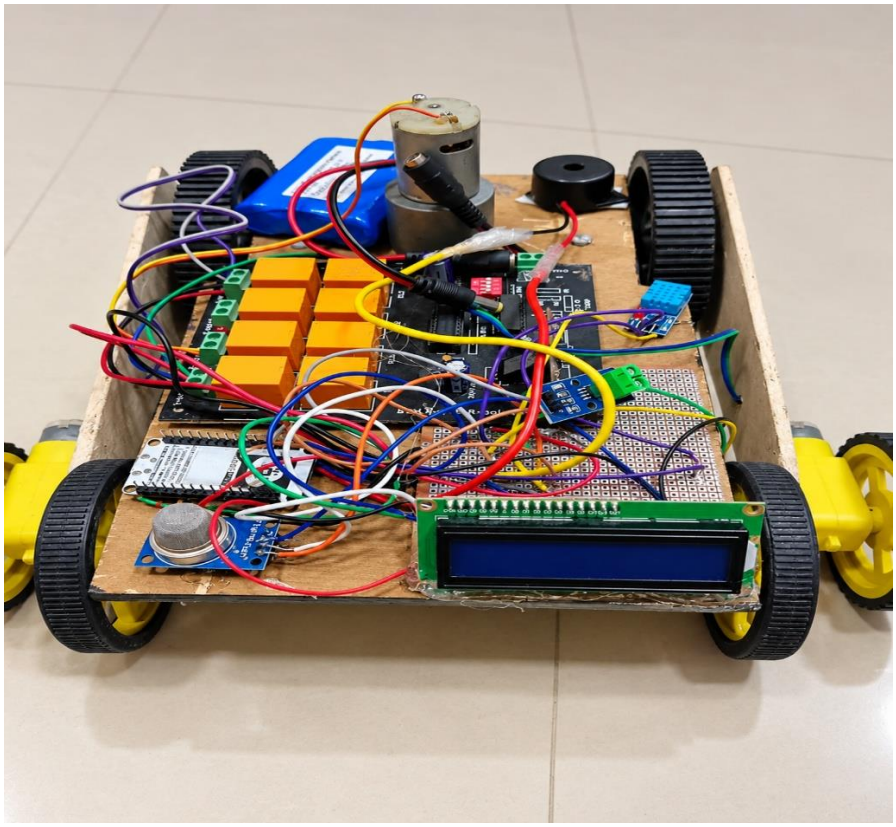
Working Principle

1. Sensors continuously collect data.
2. ESP8266 reads sensor values.
3. Data is processed and compared with threshold limits.

4. Sensor values are displayed on the LCD.
5. If abnormal conditions are detected:
    - LED turns ON.
    - Buzzer generates an alarm.
    - LCD displays warning messages.

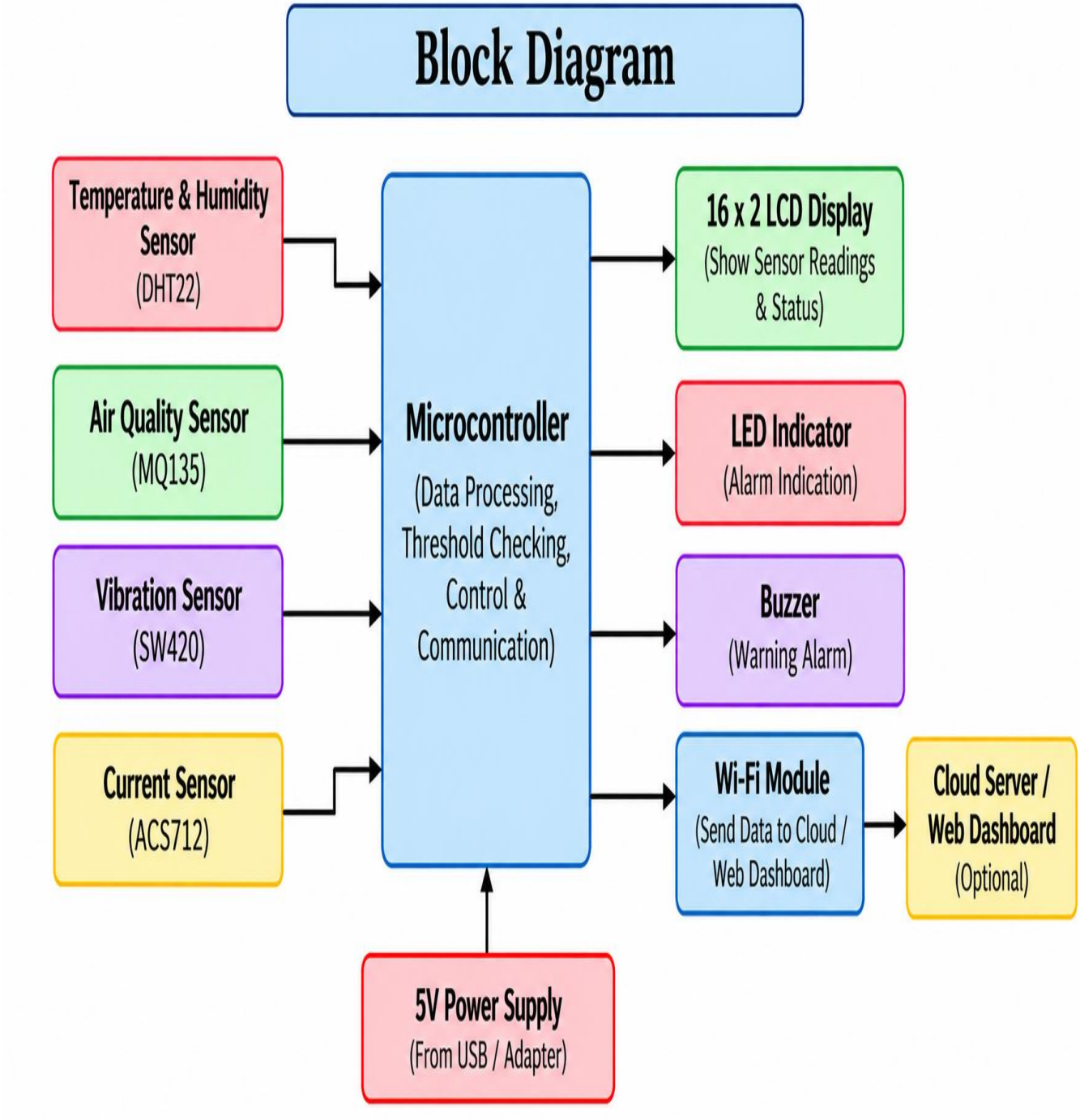


Circuit Diagram



6.3 Block Diagram

The Block Diagram represents the overall functional structure of the Industrial IoT Monitoring System. It shows how sensor data is collected, processed by the ESP8266 NodeMCU, displayed on the LCD, and used for generating warning alerts. The system continuously monitors various machine health parameters and provides real-time information to the user



Explanation of Block Diagram

The Industrial IoT Monitoring System consists of four sensing units: DHT11, MQ135, SW420, and ACS712 sensors. These sensors continuously collect information related to temperature, humidity, gas concentration, machine vibration, and current consumption. The collected data is transmitted to the ESP8266NodeMCU, which serves as the central processing unit.

The ESP8266 processes the sensor readings and displays the information on the LCD module. Whenever any sensor value exceeds its predefined threshold limit, the controller activates the LED and buzzer to generate visual and audible warning signals. The built-in Wi-Fi functionality of ESP8266 also provides support for future IoT-based remote monitoring and cloud integration.

Overall Flow:

Sensors → ESP8266NodeMCU → LCD Display → LED/Buzzer Alerts → IoT Monitoring System

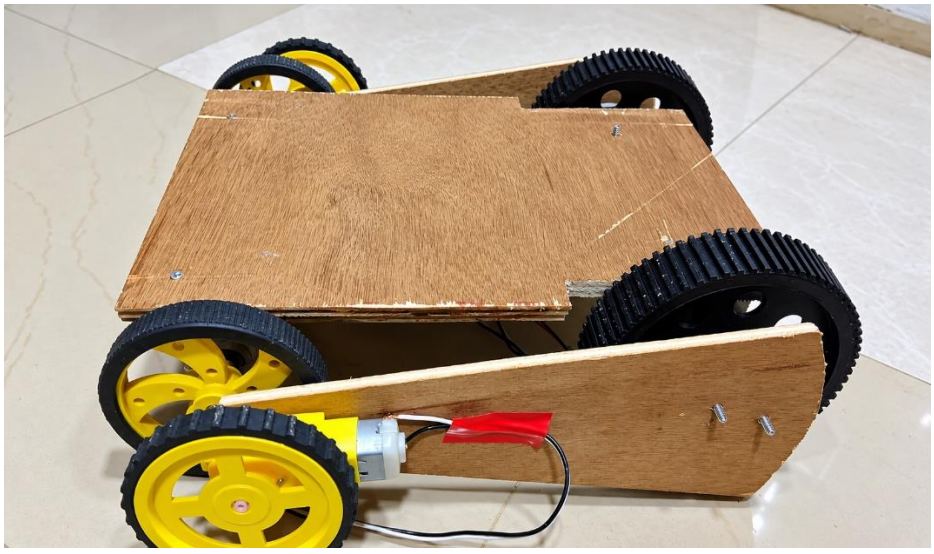
6.4 Prototype Testing Platform

The Industrial IoT Monitoring System was successfully implemented as a working prototype for testing and demonstration purposes. To facilitate easy transportation, testing, and practical demonstration, all hardware components were mounted on a mobile prototype platform.

The prototype platform serves only as a testing and demonstration setup and is not a part of the core functionality of the Industrial IoT Monitoring System. The primary objective of the project remains real-time monitoring of machine health parameters such as temperature, humidity, gas concentration, vibration, and current consumption.

The platform supports the placement of the ESP8266 NodeMCU, DHT11 Sensor, MQ135 Gas Sensor, SW420 Vibration Sensor, ACS712 Current Sensor, LCD display, LED indicator, buzzer, and power supply in a compact and organized manner. This arrangement simplifies testing, observation of sensor readings, and demonstration of the complete monitoring system.

The prototype continuously collects sensor data and displays the readings on the LCD screen. Whenever abnormal conditions are detected, the LED and buzzer generate warning alerts, demonstrating the effectiveness of the monitoring system.



FINAL REVIEW

7.1 Conclusion

The project titled **"Industrial IoT Monitoring System for Real-Time Machine Health Monitoring"** was successfully designed, developed, and tested. The primary objective of this project was to create a reliable and cost-effective monitoring system capable of continuously observing machine health parameters and generating alerts whenever abnormal conditions are detected.

The system successfully integrates the **ESP8266 NodeMCU**, **DHT11 Temperature and Humidity Sensor**, **MQ135 Gas Sensor**, **SW420 Vibration Sensor**, **ACS712 Current Sensor**, **LCD Display**, **LED**, and **Buzzer** into a single monitoring platform. These components work together to provide real-time monitoring of industrial machine conditions.

The developed system continuously measures temperature, humidity, gas concentration, vibration levels, and electrical current consumption. Whenever any monitored parameter exceeds its predefined safety limit, the system immediately generates visual and audible warnings through the LED and buzzer while displaying warning messages on the LCD screen.

The project demonstrates the practical application of Internet of Things (IoT) concepts in industrial environments. Continuous monitoring helps reduce machine downtime, improve operational efficiency, minimize maintenance costs, and increase workplace safety. The system also provides a strong foundation for predictive maintenance by enabling early identification of machine faults before they become critical failures.

The built-in Wi-Fi capability of the ESP8266 allows future expansion toward cloud-based monitoring and remote access solutions. The modular design of the system ensures that additional sensors and advanced monitoring features can be incorporated easily.

Overall Achievement Summary

The Industrial IoT Monitoring System successfully achieved the following objectives:

- Real-time machine health monitoring.
- Temperature and humidity monitoring.
- Gas leakage detection.
- Vibration monitoring.
- Current consumption monitoring.
- LCD-based data visualization.
- LED and buzzer alert generation.
- Low-cost IoT implementation.
- Expandable and scalable system architecture.

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7.2 Future Scope

Although the developed system performs efficiently, several enhancements can be incorporated in the future to improve functionality and increase industrial applicability.

Future Improvements

- Cloud-based data storage and monitoring.
- Web dashboard for remote access.
- Mobile application integration.
- SMS and Email notification system.
- Predictive maintenance using Artificial Intelligence (AI).
- Machine Learning-based fault prediction.
- Industrial database integration.
- Real-time graphical data visualization.
- Wireless remote monitoring through IoT platforms.
- GPS-based asset tracking.
- Automatic machine shutdown during critical conditions.
- Additional sensors for pressure, smoke, and sound monitoring.
-

These future improvements can transform the current system into a complete Smart Industrial Monitoring Platform suitable for Industry 4.0 applications.

7.3 References

The following books, datasheets, and technical resources were referred to during the development of this project:

1. ESP8266NodeMCU Datasheet.
2. DHT11 Temperature and Humidity Sensor Datasheet.
3. MQ135 Gas Sensor Datasheet.
4. SW420 Vibration Sensor Datasheet.
5. ACS712 Current Sensor Datasheet.
6. Arduino IDE Documentation.
7. ESP8266 Arduino Core Documentation.
8. Internet of Things (IoT) Concepts and Applications.
9. Industrial Automation and Monitoring Systems Literature.
10. Sensor Interfacing and Embedded Systems Tutorials.
11. Electronics Engineering Fundamentals.
12. Industrial Safety Monitoring Standards.
13. Embedded C Programming Guide.
14. IoT-Based Machine Health Monitoring Research Papers.
15. Online Technical Documentation and Engineering Resources.